Volume Viii Reference

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Numerical Analysis

0.1 IEEE 754 Floating-Point Formats

IEEE 754 Floating-Point Formats

Definition 0.1.1 (IEEE 754 Binary Floating-Point Datum)**.** An *IEEE 754 binary floating-point datum* is a binary encoding partitioned into three fields:

- a *sign field* consisting of one bit,
- an *exponent field* consisting of a fixed number of bits,
- a *fraction field* consisting of a fixed number of bits.

Its interpretation depends on the exponent-field pattern and the fraction-field pattern.

Definition 0.1.2 (Normalized IEEE 754 Binary Number)**.** Let an IEEE 754 binary floating-point format have exponent bias bias. A finite encoded datum is called *normalized* if its exponent field is neither all zeros nor all ones.

If s is the sign bit, if e is the unsigned integer determined by the exponent field, and if f is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s(1.f)_2 2^{e-\text{bias}}.$$

Definition 0.1.3 (Subnormal IEEE 754 Binary Number)**.** Let an IEEE 754 binary floating-point format have exponent bias bias. A finite encoded datum is called *subnormal* if its exponent field is all zeros and its fraction field is not all zeros.

If s is the sign bit and if f is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s(0.f)_2 2^{1-\text{bias}}.$$

Definition 0.1.4 (IEEE 754 Infinity Encoding)**.** An IEEE 754 binary floating-point datum encodes *infinity* if its exponent field is all ones and its fraction field is all zeros.

The sign bit determines whether the represented value is $+\infty$ or $-\infty$.

Definition 0.1.5 (IEEE 754 NaN Encoding)**.** An IEEE 754 binary floating-point datum encodes *NaN* (Not a Number) if its exponent field is all ones and its fraction field is not all zeros.

Definition 0.1.6 (IEEE 754 Binary16 Format)**.** The *IEEE 754 binary16 format* is the 16-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 5 exponent bits,
- 10 fraction bits,

- exponent bias 15.

For a normalized finite binary16 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-15}.$$

Definition 0.1.7 (IEEE 754 Binary32 Format)**.** The *IEEE 754 binary32 format* is the 32-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 8 exponent bits,
- 23 fraction bits,
- exponent bias 127.

For a normalized finite binary32 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-127}.$$

Definition 0.1.8 (IEEE 754 Binary64 Format)**.** The *IEEE 754 binary64 format* is the 64-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 11 exponent bits,
- 52 fraction bits,
- exponent bias 1023.

For a normalized finite binary64 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-1023}.$$

Definition 0.1.9 (IEEE 754 Binary128 Format)**.** The *IEEE 754 binary128 format* is the 128-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 15 exponent bits,
- 112 fraction bits,
- exponent bias 16383.

For a normalized finite binary128 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-16383}.$$

0.2 Interval Arithmetic

0.2.1 Interval Arithmetic

Definition 0.2.1 (Interval Addition)**.** For closed intervals $[a, b]$ and $[c, d]$, their *sum* is

$$[a, b] + [c, d] \ = \ [a + c, \ b + d].$$

Definition 0.2.2 (Interval Subtraction)**.** For closed intervals $[a, b]$ and $[c, d]$, their *difference* is

$$[a, b] - [c, d] \ = \ [a - d, \ b - c].$$

Definition 0.2.3 (Interval Multiplication)**.** For closed intervals $[a, b]$ and $[c, d]$, their *product* is

$$[a, b] \cdot [c, d] \ = \ \big[\min(ac, ad, bc, bd), \ \max(ac, ad, bc, bd)\big].$$

Definition 0.2.4 (Interval Containment)**.** $[a, b] \subseteq [c, d]$ if and only if $c \leq a$ and $b \leq d$.

0.2.2 Interval Arithmetic in $\mathbb{R}^n$

Definition 0.2.5 (Box / Product Interval)**.** Let $a, b \in \mathbb{R}^n$ with $a \le b$ coordinatewise (i.e., $a_i \le b_i$ for all i). The *box* (or *interval vector*) determined by (a, b) is

$$[a, b] \ := \ \{x \in \mathbb{R}^n : a \le x \le b\} \ = \ \prod_{i=1}^{n} [a_i, b_i].$$

Definition 0.2.6 (Box Containment)**.** For $a \le b$ and $c \le d$ in $\mathbb{R}^n$, we write $[a, b] \subseteq [c, d]$ iff

$$(\forall x \in \mathbb{R}^n) \ \big(x \in [a, b] \Rightarrow x \in [c, d]\big).$$

Equivalently (and most useful computationally),

$$[a, b] \subseteq [c, d] \quad \iff \quad c \le a \ \wedge \ b \le d \qquad \text{(coordinatewise)}.$$

Definition 0.2.7 (Minkowski Addition of Boxes)**.** For boxes $X = [a, b]$ and $Y = [c, d]$ in $\mathbb{R}^n$, their *Minkowski sum* is

$$X \oplus Y \ := \ \{x + y : x \in X, \ y \in Y\}.$$

For boxes, this closes in the class of boxes and satisfies the endpoint rule

$$[a, b] \oplus [c, d] \ = \ [a + c, \ b + d].$$

Definition 0.2.8 (Box Subtraction / Difference)**.** For boxes $X = [a, b]$ and $Y = [c, d]$ in $\mathbb{R}^n$, define

$$X \ominus Y \ := \ \{x - y : x \in X, \ y \in Y\}.$$

Then

$$[a, b] \ominus [c, d] \ = \ [a - d, \ b - c].$$

Definition 0.2.9 (Scalar Multiplication of a Box)**.** For $\lambda \in \mathbb{R}$ and a box $[a, b] \subseteq \mathbb{R}^n$, define

$$\lambda[a, b] \ := \ \{\lambda x : x \in [a, b]\}.$$

Then

$$\lambda[a, b] = \begin{cases} [\lambda a, \ \lambda b], & \lambda \ge 0, \\ [\lambda b, \ \lambda a], & \lambda < 0, \end{cases}$$

where the inequalities are coordinatewise.

Definition 0.2.10 (Coordinatewise / Hadamard Product of Boxes)**.** For $x, y \in \mathbb{R}^n$, write $(x \odot y)_i := x_i y_i$. For boxes $X = [a, b]$ and $Y = [c, d]$, define

$$X \odot Y \ := \ \{x \odot y : x \in X, \ y \in Y\}.$$

Then $X \odot Y$ is again a box, computed coordinatewise:

$$[a, b] \odot [c, d] \ = \ \prod_{i=1}^{n} \Big[\min S_i, \ \max S_i\Big], \qquad S_i := \{a_i c_i, \ a_i d_i, \ b_i c_i, \ b_i d_i\}.$$

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Convexity and Hulls

Definition 0.13.1 (Convex Set)**.** A set $S \subseteq \mathbb{R}^2$ is *convex* if for every pair of points $a, b \in S$ and every $\lambda \in [0, 1]$,

$$\lambda a + (1 - \lambda)b \in S.$$

Equivalently, S contains the line segment between any two of its points.

Definition 0.13.2 (Convex Combination)**.** A *convex combination* of points $q_1, \dots, q_n \in \mathbb{R}^2$ is a point of the form

$$\sum_{i=1}^{n} \lambda_i q_i, \qquad \lambda_i \geq 0 \text{ for all } i, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

The numbers λ_i are called the weights.

Definition 0.13.3 (Convex Hull)**.** The *convex hull* of a set $P \subseteq \mathbb{R}^2$, written $\operatorname{conv}(P)$, is the set of all convex combinations of finitely many points of P:

$$\operatorname{conv}(P) = \left\{ \sum_{i=1}^{n} \lambda_i q_i \;\middle|\; n \geq 1,\ q_i \in P,\ \lambda_i \geq 0,\ \sum_{i=1}^{n} \lambda_i = 1 \right\}.$$

Equivalently, $\operatorname{conv}(P)$ is the smallest convex set containing P, i.e. the intersection of all convex sets that contain P.

Lemma 0.13.4 (Convex hull as the smallest convex set)**.** *Let $P \subseteq \mathbb{R}^2$. Then $\operatorname{conv}(P)$ is convex, contains P, and is contained in every convex set that contains P. Hence*

$$\operatorname{conv}(P) = \bigcap \{K \subseteq \mathbb{R}^2 : K \textit{ is convex and } P \subseteq K\}.$$

Supporting Functionals

Definition 0.13.5 (Extreme Point)**.** Let $S \subseteq \mathbb{R}^2$ be convex. A point $p \in S$ is an *extreme point* of S if whenever

$$p = \lambda a + (1 - \lambda)b, \qquad a, b \in S, \quad \lambda \in (0, 1),$$

it follows that $a = b = p$. For a finite point set P, the extreme points of $\operatorname{conv}(P)$ are called the vertices of the convex hull.

Definition 0.13.6 (Supporting Line)**.** A line $\ell \subseteq \mathbb{R}^2$ is a *supporting line* of a convex set S at a point $p \in S$ if $p \in \ell$ and S lies entirely in one of the two closed half-planes bounded by ℓ. Equivalently, there exists $d \in \mathbb{R}^2 \setminus \{0\}$ such that

$$\ell = \{x \in \mathbb{R}^2 : \langle d, x \rangle = \langle d, p \rangle\}$$

and

$$\langle d, q \rangle \leq \langle d, p \rangle \qquad \text{for all } q \in S.$$

Lemma 0.13.7 (Linear functionals preserve convex combinations)**.** *Let $d \in \mathbb{R}^2$, and let*

$$z = \sum_{i=1}^{n} \lambda_i q_i$$

be a convex combination of points $q_1, \ldots, q_n \in \mathbb{R}^2$. Then

$$\langle d, z \rangle = \sum_{i=1}^{n} \lambda_i \langle d, q_i \rangle.$$

Consequently, if $\langle d, q_i \rangle \le M$ for every i, then

$$\langle d, z \rangle \le M.$$

Theorem 0.13.8 (Supporting functional characterization of hull vertices)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. A point $p \in P$ is a vertex of $\operatorname{conv}(P)$ if and only if there exists a direction $d \in \mathbb{R}^2 \setminus \{0\}$ such that*

$$\langle d, p \rangle > \langle d, q \rangle \qquad \text{for all } q \in P \setminus \{p\}.$$

Equivalently, p is the unique point of P attaining

$$\max_{q \in P} \langle d, q \rangle.$$

Thus every hull vertex is a maximum of P under some linear functional.

Upper Hull Structure

Theorem 0.13.9 (Monotone slope characterization of the upper hull)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. Let*

$$v_0, v_1, \ldots, v_m$$

be the vertices of the upper hull of P, listed in increasing x-coordinate, and define

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \qquad k = 0, 1, \ldots, m-1.$$

Then

$$s_0 > s_1 > \cdots > s_{m-1}.$$

Conversely, if points $w_0, \ldots, w_m$ satisfy

$$x(w_0) < x(w_1) < \cdots < x(w_m)$$

and their consecutive slopes are strictly decreasing, then every w_k is a vertex of

$$\operatorname{conv}(\{w_0, \ldots, w_m\}).$$

The lower hull is characterized similarly, with strictly increasing consecutive slopes.

Theorem 0.13.10 (Greedy slope construction of the upper hull)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. Define*

$$v_0 = \arg\min_{p \in P} x(p).$$

Given v_k with

$$x(v_k) < \max_{p \in P} x(p),$$

define

$$v_{k+1} = \mathop{\arg\max}_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

Terminate once

$$v_k = \arg\max_{p \in P} x(p).$$

Then the points

$$v_0, v_1, \ldots, v_m$$

produced by this construction are exactly the vertices of the upper hull of P, and the slopes encountered along the way are strictly decreasing.

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Definition 0.28.1 (Differential Equation)**.** A **differential equation** is an equation whose unknown is a function and whose terms involve one or more derivatives of that function.
For an unknown function $y = y(t)$, a differential equation may be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0.$$

Here t is the independent variable, $y(t)$ is the unknown function, and

$$y', y'', \dots, y^{(n)}$$

are derivatives of y.

Definition 0.28.2 (Order of a Differential Equation)**.** The **order** of a differential equation is the highest derivative of the unknown function that appears in the equation.
For example, if an equation involving an unknown function $y = y(t)$ can be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0$$

and $y^{(n)}$ is the highest derivative that appears, then the differential equation has **order** n.

Definition 0.28.3 (Solution of a Differential Equation)**.** A **solution** of a differential equation is a function that satisfies the equation on a specified interval.
More precisely, if a differential equation is written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0,$$

then a solution on an interval I is a function $y : I \to \mathbb{R}$ such that y has all derivatives required by the equation and

$$F(t, y(t), y'(t), y''(t), \dots, y^{(n)}(t)) = 0$$

for every $t \in I$.

Definition 0.28.4 (Autonomous Differential Equation)**.** An **autonomous differential equation** is a differential equation in which the independent variable does not appear explicitly.
For a first-order ordinary differential equation, an autonomous equation has the form

$$y' = f(y),$$

where the rate of change of y depends only on the current value of y, not directly on time t.

Definition 0.28.5 (Non-Autonomous Differential Equation)**.** A **non-autonomous differential equation** is a differential equation in which the independent variable appears explicitly.

For a first-order ordinary differential equation, a non-autonomous equation has the form

$$y' = f(t, y),$$

where the rate of change of y may depend both on the current value of y and on the independent variable t.

Definition 0.28.6 (Linear Differential Equation)**.** A differential equation is called **linear** if the unknown function and its derivatives appear only to the first power and are not multiplied together.

An nth-order linear ordinary differential equation for an unknown function $y = y(t)$ has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

where the coefficient functions

$$a_0(t), a_1(t), \ldots, a_n(t)$$

and the forcing term $g(t)$ are given functions of the independent variable t.

Definition 0.28.7 (Malthusian Population Growth Model)**.** The **Malthusian population growth model** assumes that the rate of change of a population is proportional to the current population.

If $P(t)$ denotes the population at time t, then the model is

$$P'(t) = kP(t),$$

where k is a constant called the **growth rate** or **proportionality constant**.

Definition 0.28.8 (Decay)**.** A quantity is said to undergo **decay** if it decreases over time.

In a differential equation model, if $Q(t)$ denotes the quantity at time t, then decay is often modeled by an equation of the form

$$Q'(t) = -kQ(t),$$

where $k > 0$ is a constant.

Definition 0.28.9 (Newton's Law of Cooling)**.** **Newton's Law of Cooling** states that the rate at which an object's temperature changes is proportional to the difference between the object's temperature and the surrounding ambient temperature.

If $T(t)$ denotes the temperature of the object at time t, and T_a denotes the ambient temperature, then the model is

$$T'(t) = -k\big(T(t) - T_a\big),$$

where $k > 0$ is a constant called the **cooling constant**.

Definition 0.28.10 (General Solution)**.** A **general solution** of a differential equation is a family of solutions containing one or more arbitrary constants.

For an nth-order ordinary differential equation, the general solution typically contains n arbitrary constants:

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n).$$

Each choice of the constants $C_1, C_2, \ldots, C_n$ gives a particular solution of the differential equation.

Definition 0.28.11 (Family of Solutions)**.** A **family of solutions** of a differential equation is a collection of solution functions described by one or more parameters.

For example, a family of solutions may be written in the form

$$y(t) = \Phi(t, C),$$

or more generally,

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n),$$

where $C, C_1, \ldots, C_n$ are parameters. Each choice of the parameters gives one particular solution.

Definition 0.28.12 (Initial Condition)**.** An **initial condition** specifies the value of the unknown function at a particular value of the independent variable.
For a first-order differential equation involving an unknown function $y = y(t)$, an initial condition has the form

$$y(t_0) = y_0,$$

where t_0 is the initial time and y_0 is the initial value.

Definition 0.28.13 (Initial Value Problem)**.** An **initial value problem**, or **IVP**, is a differential equation together with one or more initial conditions.
For a first-order differential equation, an initial value problem has the form

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

The differential equation describes how the function changes, while the initial condition specifies the value of the solution at the initial time t_0.

Definition 0.28.14 (Existence)**.** An initial value problem is said to have **existence** if there is at least one function that satisfies both the differential equation and the initial condition.
For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

existence means that there is a function $y(t)$, defined on some interval containing t_0, such that

$$y'(t) = f(t, y(t))$$

and

$$y(t_0) = y_0.$$

Definition 0.28.15 (Uniqueness)**.** An initial value problem is said to have **uniqueness** if it has at most one solution on the interval under consideration.
For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

uniqueness means that if $y_1(t)$ and $y_2(t)$ are both solutions on the same interval I containing t_0, then

$$y_1(t) = y_2(t)$$

for every $t \in I$.

Definition 0.28.16 (Interval of Existence)**.** An **interval of existence** for a solution of an initial value problem is an interval on which the solution is defined and satisfies the differential equation.
For an initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

an interval I is an interval of existence if $t_0 \in I$ and there is a solution $y : I \to \mathbb{R}$ such that

$$y'(t) = f(t, y(t))$$

for every $t \in I$, and

$$y(t_0) = y_0.$$

Proofs

Partial Differential Equations

0.29 Partial Differential Equations

Proofs

Computational Probability

0.30 Computational Probability

Proofs

Applied and Computational Mathematics